

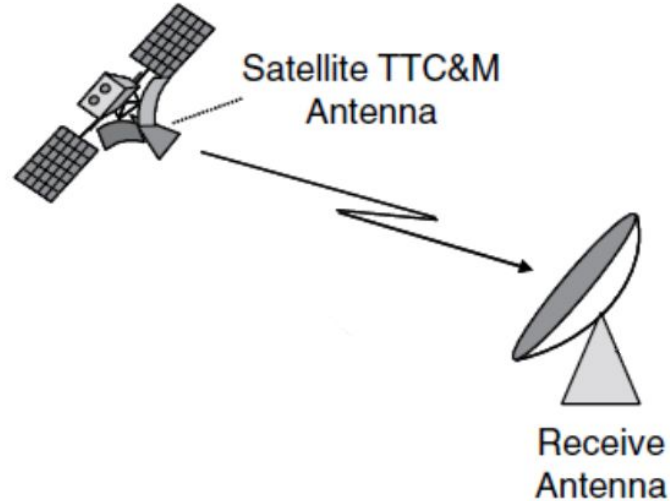
# Let's learn about **satellites!**

**Space mission** and **satellite** design course

## **Satellite communication System**

# Telecommunication

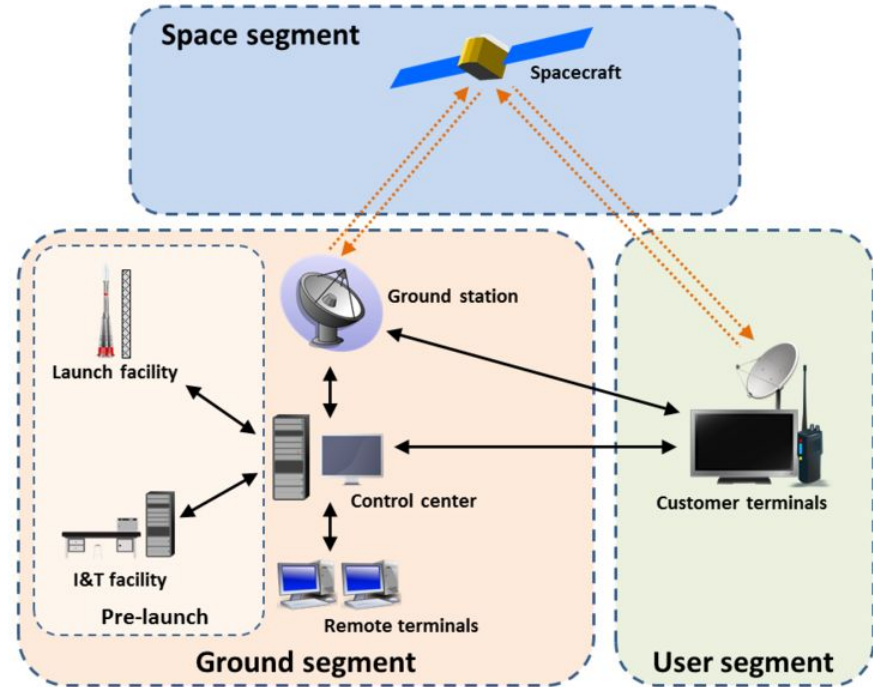
The telecommunication system is critical for a space mission. With no telecommunication the spacecraft is simply a very expensive **debris**.



# Segments of a space mission

A spacecraft mission is divided into three segments:

- Space
- Ground
- User



# Why do we communicate?

- **Telemetry:** satellites must send the health status of its systems to the ground
- **Commands:** ground must send commands to the spacecraft in order to actively perform the mission
- **Tracking:** radio frequencies used for the communication can be used to know position and velocity of the satellite itself
- **Payload:** the satellite gathers data and sends them back to the ground. (e.g. planet magnetosphere)

# Mission data size

How much data generates each part of the mission?

1. **Telemetry:** typically ranges between **bytes** and **kbytes**
2. **Commands:** typically ranges between **bytes** and **Mbytes**
3. **Payload:** typically ranges between **kbytes** and **Tbytes**

# Mission example

Let's take, for instance, a 1U mission which has a **4MP** payload camera.

- We take **150** photos a day.
- The telemetry parameters are read one time every **5** seconds

# Mission example - 2

We imagine to have the following telemetry parameters:

- X,Y,Z position: 24 bytes - X,Y,Z rotation: 24 bytes
- 10 temperature sensors: 1 byte x 10 = 10 bytes
- 10 voltage sensors: 1 byte x 10 = 10 bytes
- 10 current sensors: 1 byte x 10 = 10 bytes
- 64 On/Off variables: 8 bytes
- 20 other parameters: 4 bytes x 20 = 80 bytes

Total bytes: **166** bytes

Total data per day:  $1223 * 86400 / 5 = \mathbf{2.8}$  Mbytes/day

# Mission example - 3

A 4MP camera produces 3 bytes of data for each pixels which generates **12Mbytes** of data for each photo.

Using a lossless compression with a ratio of 2:1 which reduces the picture weight to **6Mbytes**.

**6Mbytes** x 150 pic/day = 900 Mbytes/day



# Mission example - 4

- **Telemetry:** 2.8 Mbytes/day
- **Payload:** 900 Mbytes/day

Telemetry is only **0.32%** of the data generated!

This example tells us that, almost every time, the design driver of the telecommunication system is the payload.

# Telecommunication design

In order to design a satellite telecommunication system:

1. We must identify the **requirements** of the mission;
2. Define the **architecture** (e.g. number of ground stations)
3. Identify the **data rate** required (bit per second or bps)
4. Link **sizing**
5. **Components** choice

# Telecommunication design - 2

The main requirements are:

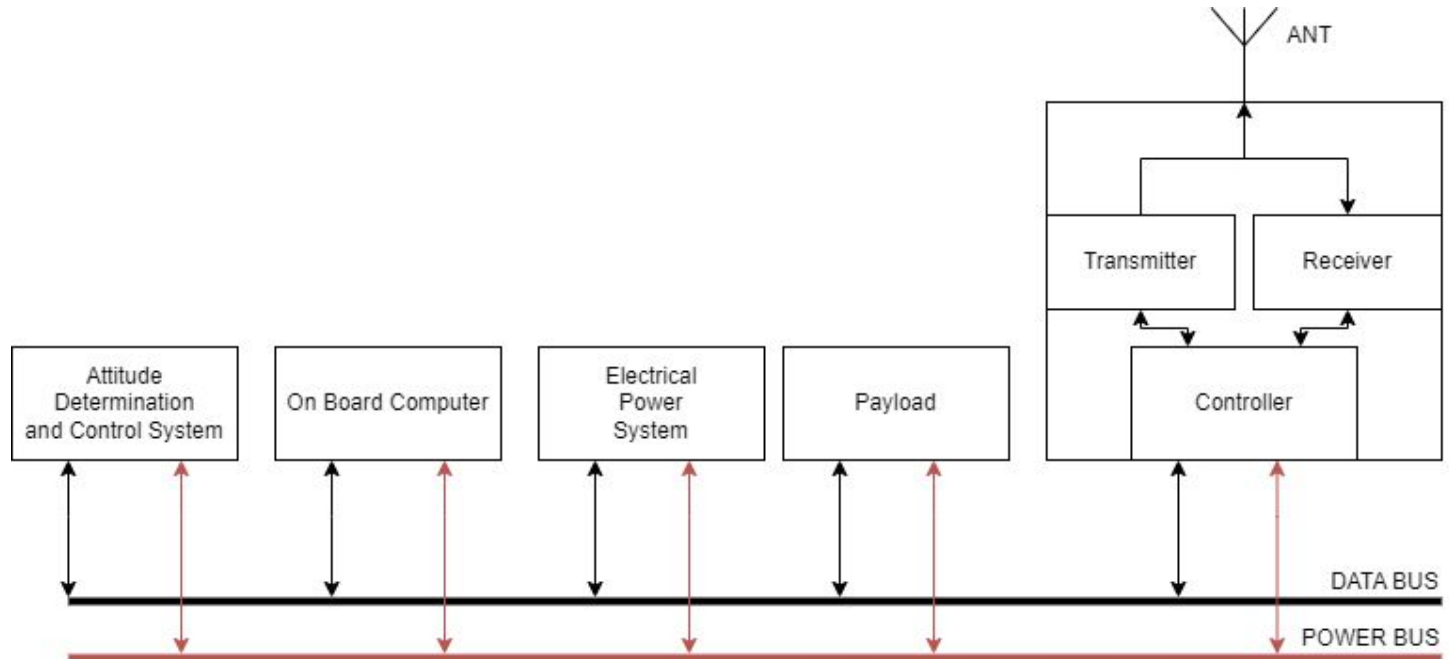
- **Data volume:** the amount of information to transmit and receive.
- **Timing/Coverage:** when the system will be active, which frequency to use for **downlink** and **uplink**.
- **Link quality:** service quality, availability, reliability

The main **limitations** are:

**technical** limitations, **environmental** constraints, international **regulations**

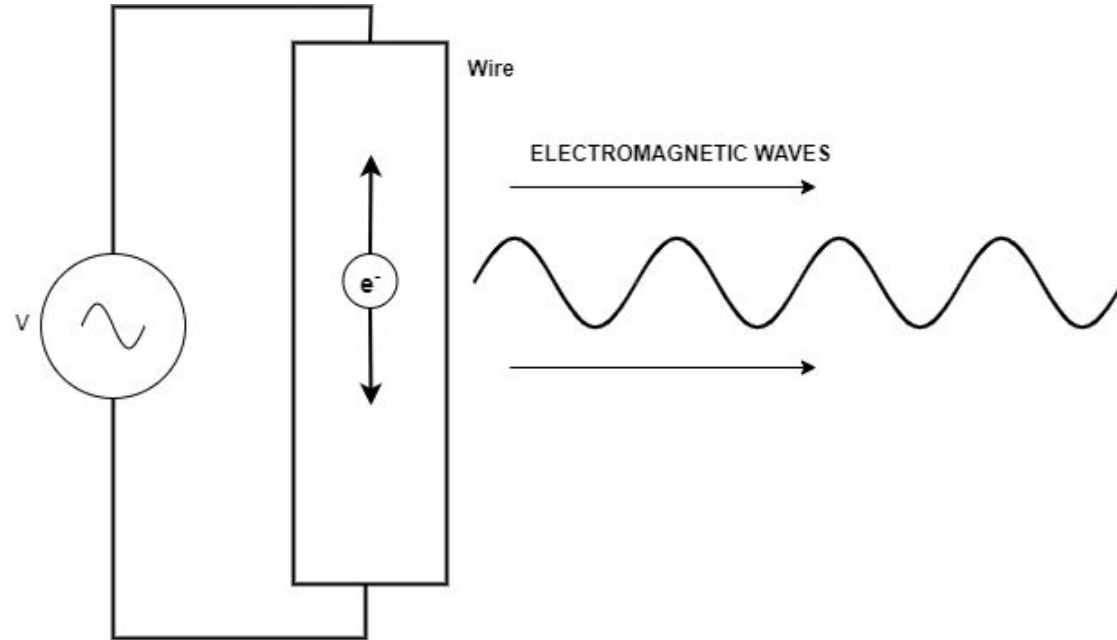
# Telecommunication design - 3

Example of how the telecommunication system fit in the spacecraft



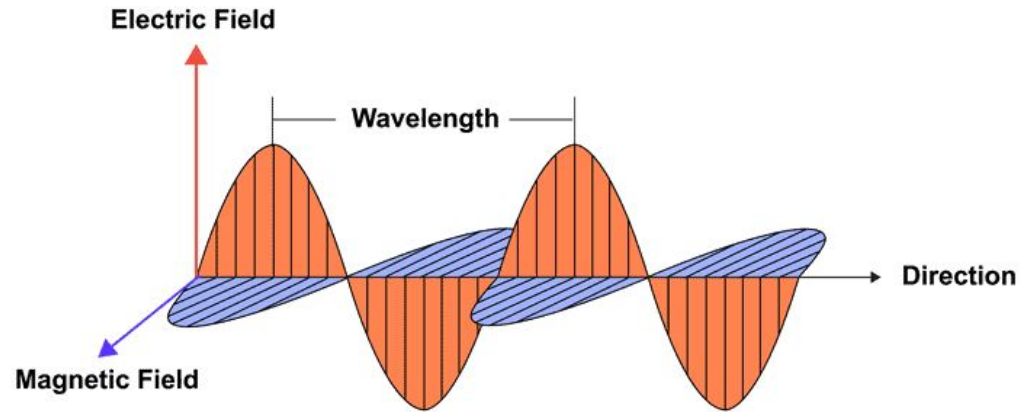
# Electromagnetic wave - Creation

How are the electromagnetic waves created in order to communicate



# Electromagnetic wave

Electromagnetic waves are used to transmit information wireless and they are composed by the electric field and magnetic field orthogonal to each other



**Electromagnetic Wave**

# Frequencies

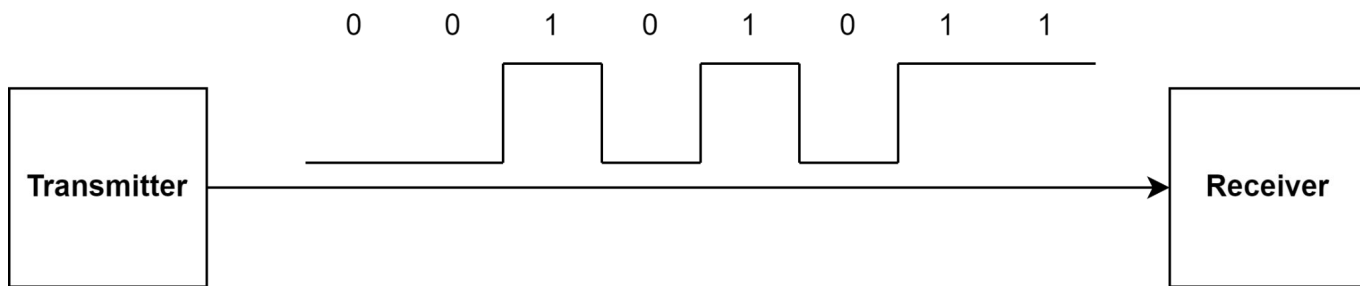
|     | Band           |                                                                                        |
|-----|----------------|----------------------------------------------------------------------------------------|
| VHF | 30 - 300 MHz   | Typically used by missions with low data volume or for telemetry downlink              |
| UHF | 300 - 1000 MHz |                                                                                        |
| L   | 1 - 2 GHz      | Used for higher data volume missions, GEO and interplanetary missions                  |
| S   | 2 - 4 GHz      |                                                                                        |
| C   | 4 - 8 GHz      |                                                                                        |
| X   | 8 - 12 GHz     | Used by interplanetary missions or LEO missions with very high data volume to transmit |
| Ku  | 12 - 18 GHz    |                                                                                        |
| K   | 18 - 27 GHz    |                                                                                        |
| Ka  | 27 - 40 GHz    |                                                                                        |

# Digital data transmission

Almost every data is now transmitted in **digital** form. This means that we have to transmit only ones or zeros.

This information can be interpreted electrically by different voltages.

We represent a **1** using **5V** and we represent a **0** using **0V**.





# Modulation

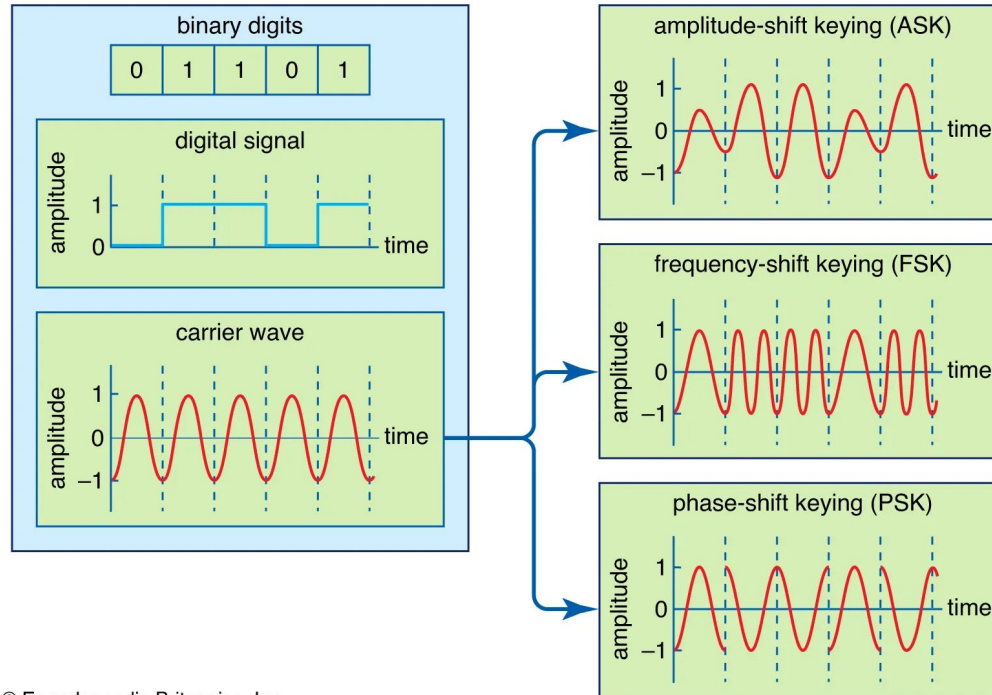
In order to be able to communicate, we need to merge two signals:

1. **Digital signal**
2. **Carrier wave**

The output is the **modulated** signal. The three main type of modulation are:

- **ASK (Amplitude-Shift Keying)**
- **FSK (Frequency-Shift Keying)**
- **PSK (Phase-Shift Keying)**

# Modulation - 2



# Modulation - 3

The type of modulation is important because each one of them gives us different peculiarities such as:

- **Bandwidth**: typically higher bandwidth means higher data rates;
- **Noise resistance**: how many bits we receive are corrupted respect to the noise present in the channel;
- **Technological difficulties**: some modulations are more difficult to obtain hence more expensive than others;

# Antennas

The modulated signal is transmitted and received using an **antenna**

The basic parameters of an antenna are:

1. **Gain (directionality)**
2. **Central frequency**
3. **Bandwidth**
4. **Radiation pattern**
5. **Polarization**
6. **VSWR**

# Monopole antenna

Monopole antenna is good for transmitting in every direction. This means that the gain is **~0dBi**

- The length of the beam can be a fraction of required wavelength (typically used in cubesat  $\frac{1}{4}$ )

300'000'000m/s : 433 MHz = 69cm (17cm)

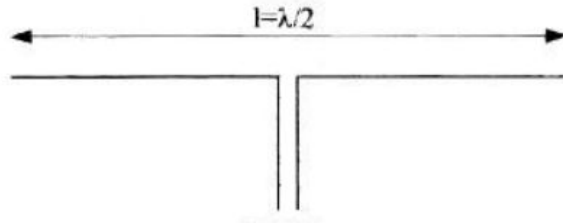


# Dipole antenna

The dipole antenna has a gain of **2.15dB**.

It can transmit in every direction except for the direction of the wire extension.

- The length of the single beam can be a fraction of required wavelength (typically used in cubesat  $\frac{1}{4}$ )
- Geometry of the beams aperture influences the impedance of the dipole



# Parabolic antenna

The parabolic antenna works for better at high frequencies ( > 2GHz).

The gain of the parabola is a function of the **frequency** and **diameter** of the parabola reflector.

$$G = 20 \log \left( \frac{\pi D f}{c} \right)$$

e.g. 8.2GHz and 8m diameter -> **56.7dB**



# Cubesat dipole antenna

Most common for low frequency antennas. The maximum theoretical gain is 2dBi

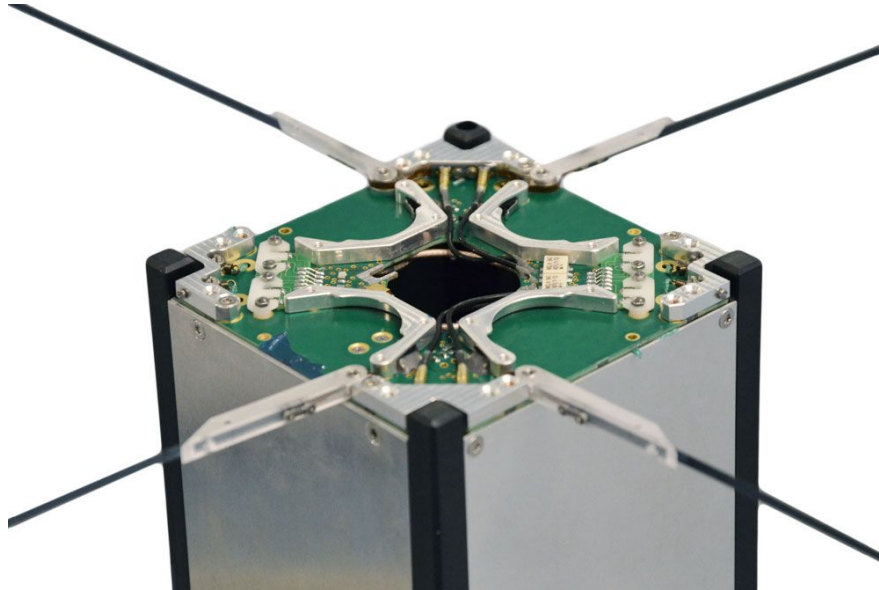


KySat-1



# Cubesat turnstile antenna

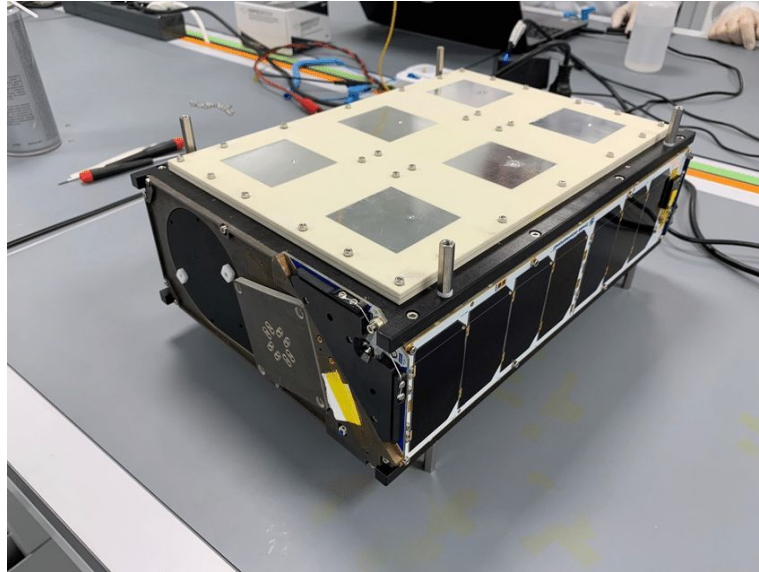
Can achieve **circular polarization** increasing the resistance to noise.



ISISpace antenna

# Cubesat patch antenna

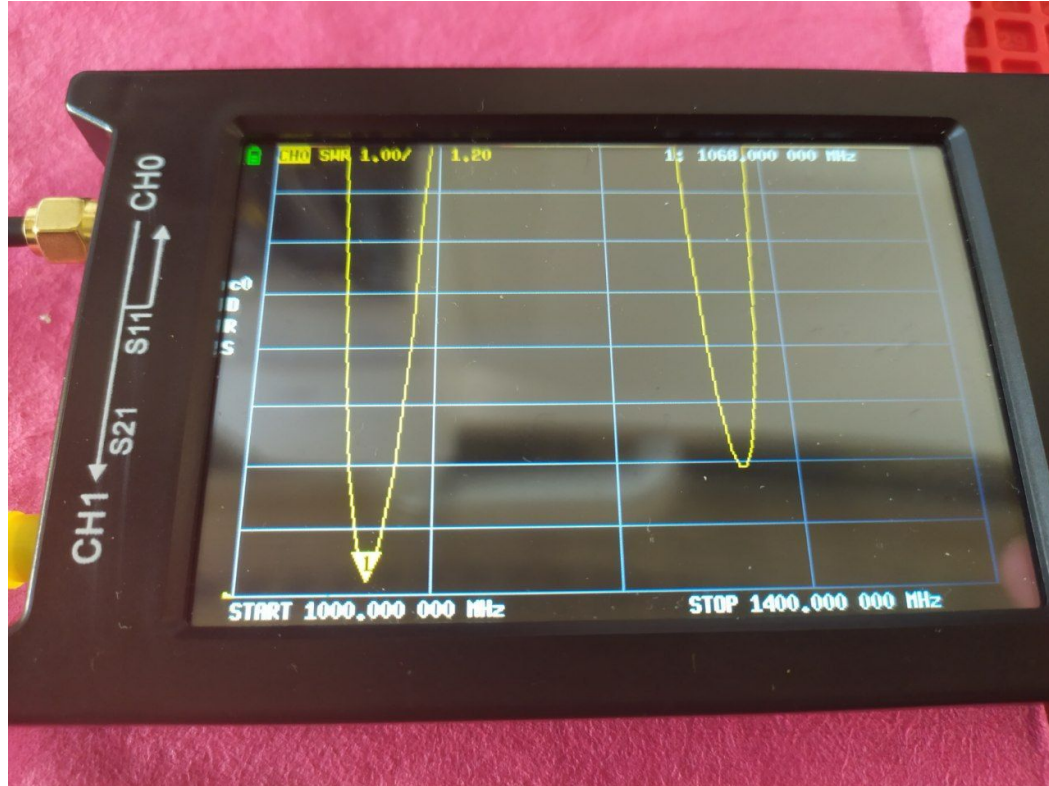
High frequency, high gain antennas (Tyvak's 6U CubeSat platform)



# Cubesat patch antenna - 2



# Cubesat patch antenna - 3



# Decibel

When we are talking about very large or very small numbers it is convenient to use the **decibels**.

$$N \text{ [dB]} = 10 \log_{10} \left( \frac{N}{N_{\text{ref}}} \right)$$

e.g.:

- 1W = 0dBW = 30 dBmW
- 10W = 10dBW = 40dBmW
- 100W = 20dBW
- 1000W = 30dBW
- 10,000W = 40dBW

# Decibel - 2

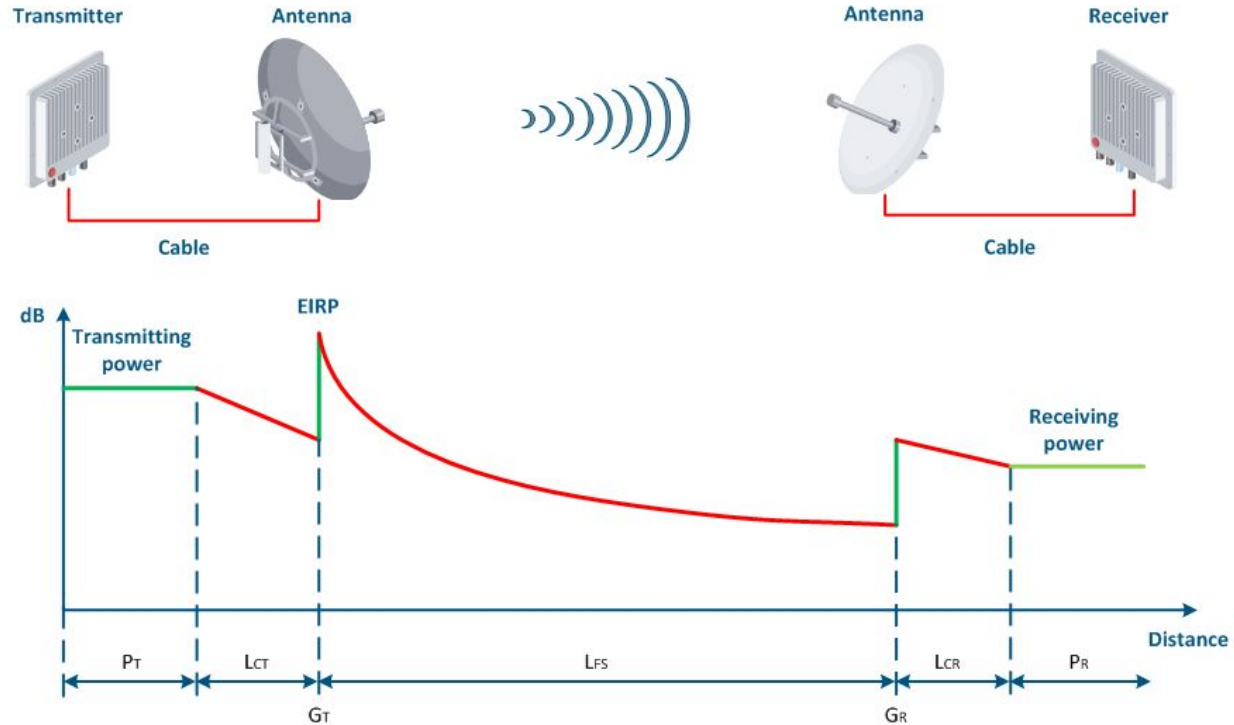
If we want go back from decibel to the number:

$$N = N_{\text{ref}} 10^{\frac{N[\text{dB}]}{10}}$$

e.g.:

- 0dB = 1
- 3dB = 2
- -3dB = 0.5
- -75dBm = 3.1e-8 mW

# Link budget



# Cable losses - example RG174

A realistic length for cubesat application is 15cm. If we are communicating at 433MHz; the losses are:

$$54.90 \text{ dB} / 100\text{m} * 0.15\text{m} = \mathbf{0.08 \text{ dB}}$$

For a ground station:

$$54.90\text{dB} / 100\text{m} * 2\text{m} = \mathbf{1.1 \text{ dB}}$$

Attenuation dB/100 m.

|          |       |
|----------|-------|
| 50 MHz   | 17,50 |
| 100 MHz  | 25,80 |
| 200 MHz  | 38,20 |
| 400 MHz  | 54,90 |
| 800 MHz  | 77,00 |
| 1000 MHz | 87,50 |

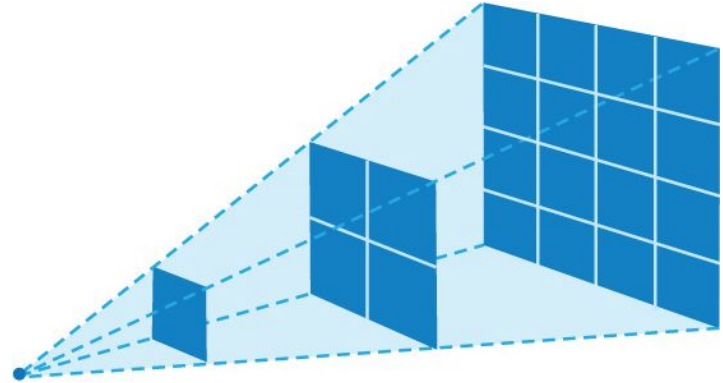
RG174 Datasheet



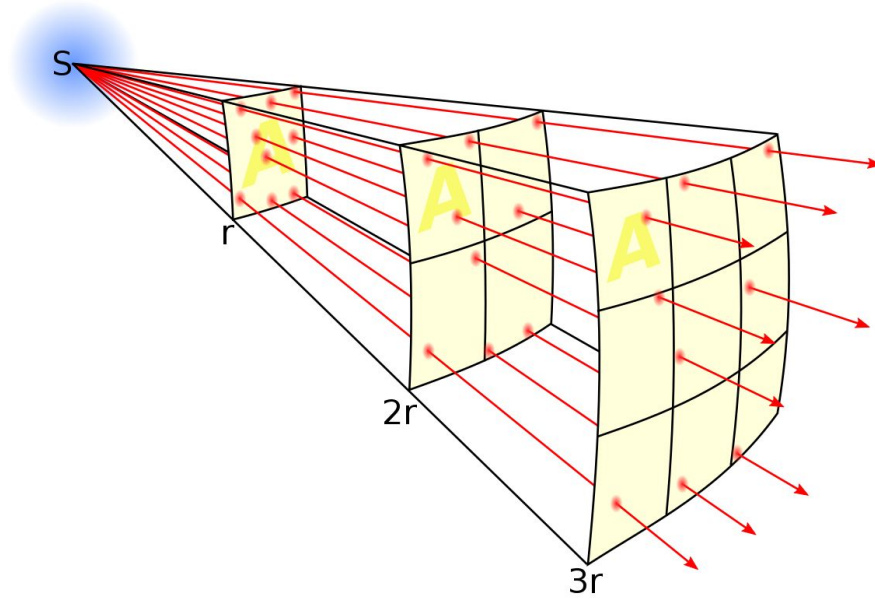
# Free space losses

The equation for calculating the **Free Space Losses**:

$$\text{FSL} = 10 \log_{10} \left( \left( \frac{4 \pi d f}{c} \right)^2 \right)$$



# Free space losses - 2



# Why higher frequencies?

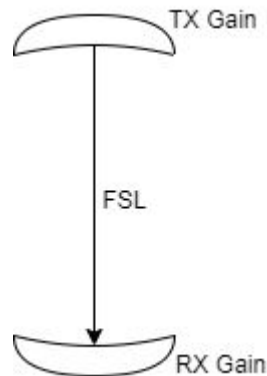
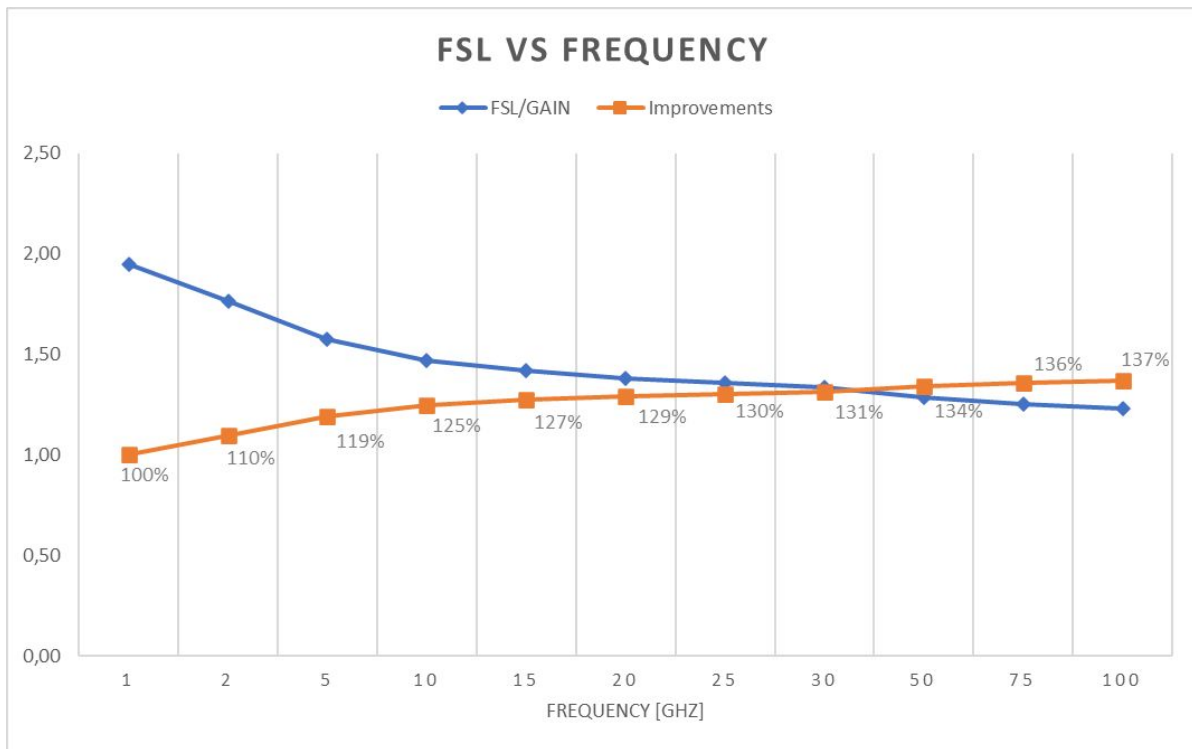
Why do we increase frequencies as we increase data rate?

1. To transmit more data we need more **bandwidth**. [430-440 MHz UHF amateur band, typical 8-PSK at 25Mbps has bandwidth > 20MHz]
2. With higher frequencies we are able to construct more sophisticated antennas with more gain

But... we have seen that if we increase the frequency we increase the free space losses. Is it worth it to increase the frequency?

Let's take a simple example with increasing frequency with two parabolic antennas with fixed and equal diameter.

# Why higher frequencies? - 2



# Other losses

- Polarization losses
- Pointing losses
- Atmospheric losses
- System temperature noise

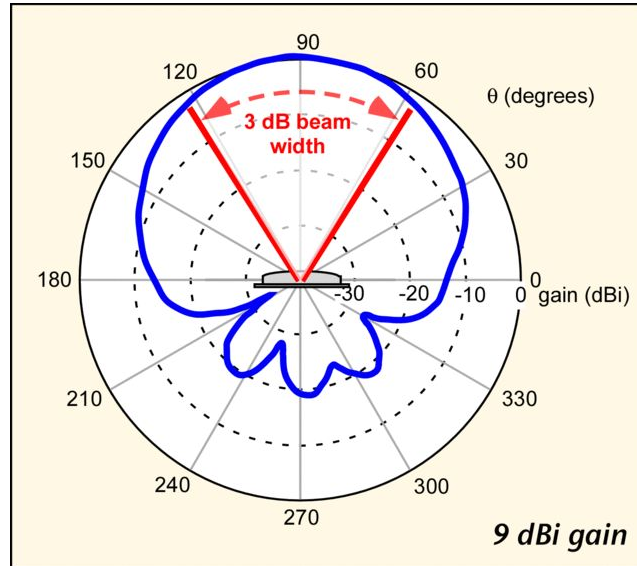
# Polarization losses

Losses due to different polarization between transmitter and receiver antenna

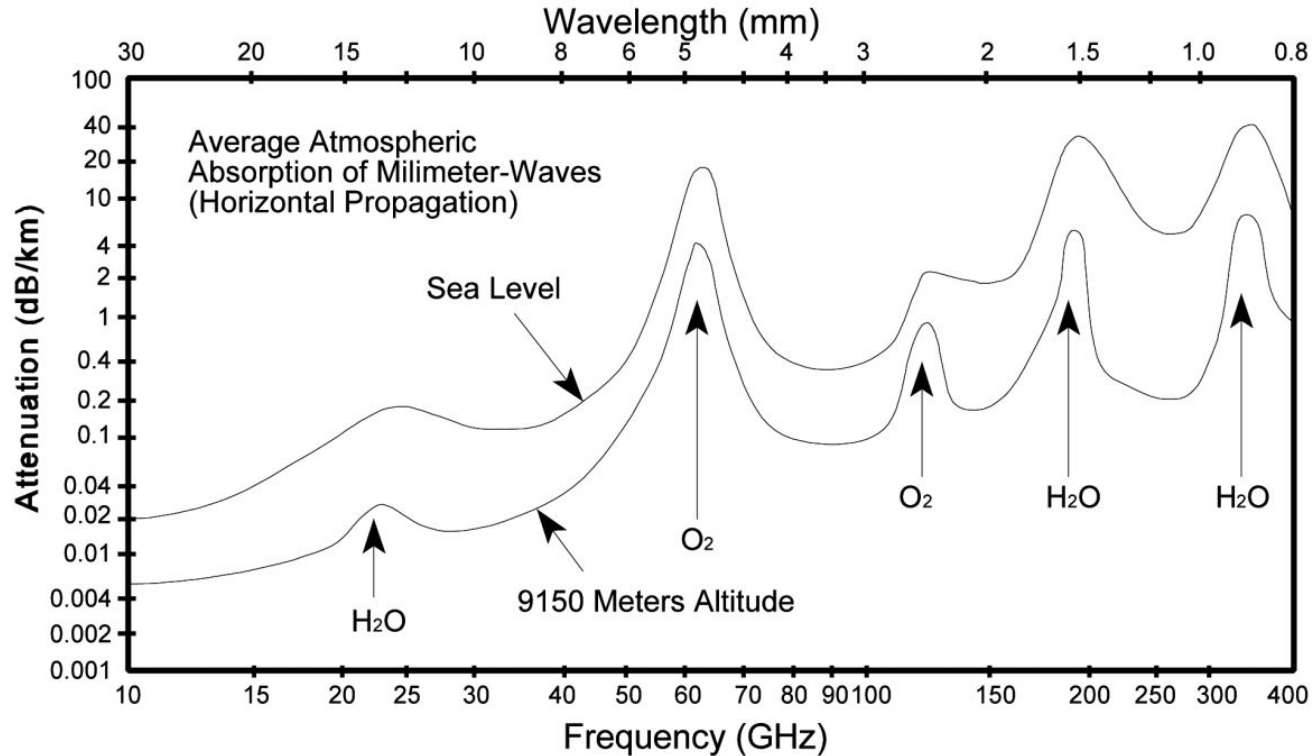
| Polarization combination                  | Polarization loss (dB) |
|-------------------------------------------|------------------------|
| Vertical - Vertical                       | 0                      |
| Vertical - Horizontal                     | $-\infty$              |
| Horizontal - Horizontal                   | 0                      |
| Circular (right or left) - Linear         | -3                     |
| Circular right hand - Circular right hand | 0                      |
| Circular right hand - Circular left hand  | $-\infty$              |

# Pointing losses

Both the antennas are not pointed exactly through the line of maximum gain

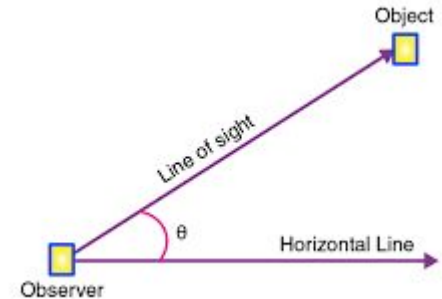
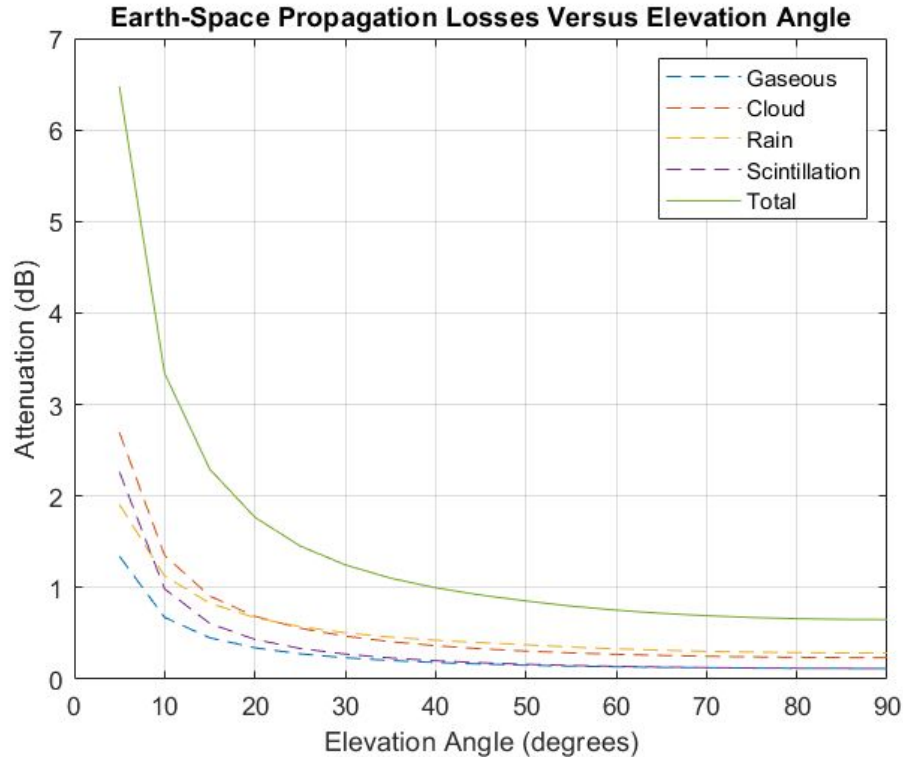


# Atmospheric losses





# Atmospheric losses - 2



# System temperature Noise

All the noise that is inserted by the mechanical and electrical system on the signal is measured in kelvin.

Typical values are:

1. 300K = 24.7dB (for DOWNLINK)
2. 600K = 27.8dB (for UPLINK)

# Bit Error Rate

How much power do we need to receive in order to have a good link quality?

While the stream of bits is arriving, we will have a certain amount of bits that are **correct** or **corrupted** depending on the amount of noise power respect to the signal power.

If the **noise power** increases with a constant **signal power**, we are going to have an increase in wrong bits (we read 0 instead of 1 and vice-versa)

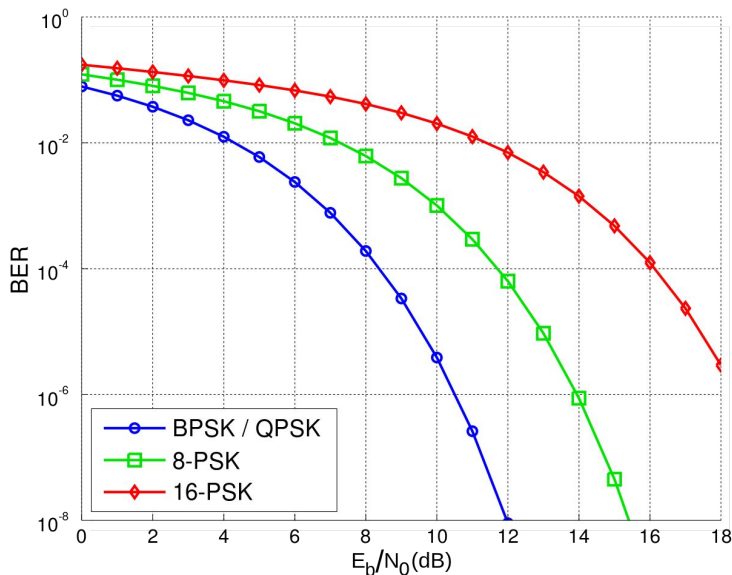
We can define the bit error rate as the **probability** to **read incorrectly** a bit.

If the B.E.R. is **100%** means that all the bits are wrong

If the B.E.R. is **0%** means that all the bits are correct. (what we would like to have)

# Bit Error Rate [BER]

The probability to have a **corrupted** bit is a function of the power transmitted over the noise power ratio; and it is called **EbN0**



e.g. If we have a **BER** of  $10^{-4}$  it means that 1 bit every 10,000 is wrong.

To further decrease the BER we can use **Error detection and correction algorithms**

# Link Budget Equation

The link budget equation gives us the maximum possible data rate we can achieve with the design parameters

$$\text{Data rate [dB]} = P_T + G_T - \text{FSL} - L_{\text{ATM}} - L_{\text{tot}} + G_R + 228.6 - \text{EbN0}$$

If the data rate is fixed we can use the following equation.

$$P_T + G_T - \text{FSL} - L_{\text{ATM}} - L_{\text{tot}} + G_R + 228.6 - \text{EbN0} - R > 0$$

# Example 1 - Data rate calculation

- We have a **3U** cubesat mission in **LEO**.
- We are using a dipole antenna at 433MHz with **2dB** gain.
- The ground station has a yagi antenna with **12dB** gain.
- We are transmitting at **0.5W** with negligible cable losses for the s/c but with **1dB** loss for the receiver
- The link distance is **900 km**
- We communicate with BPSK modulation with  **$10^{-6}$**  BER
- Atmospheric, pointing and polarization losses are negligible

# Example 1 - FSL

The equation for calculating the **Free Space Losses**:

$$\text{FSL} = 10 \log_{10} \left( \left( \frac{4 \pi d f}{c} \right)^2 \right) = 144.3 \text{dB}$$

- d (distance) = 900 km
- f (frequency) = 433 MHz
- c (speed of light) = 300,000 km/s

# Example 1 - Data rate calculation

|         |              |   |    |                   |      |
|---------|--------------|---|----|-------------------|------|
| Pt      | -3.01        | + | dB | Transmitted power |      |
| Gt      | 2            | - | dB | Transmitter gain  |      |
| FSL     | 144.3        | - | dB | Free space losses |      |
| L cable | 1            | + | dB | Cable losses      |      |
| Gr      | 12           | - | dB | Receiver gain     |      |
| T       | 24.8         | + | dB | System noise      |      |
|         | 228.6        | - | dB | Boltzman Const.   |      |
| EbN0    | 12.0         | = | dB | BER 1E-6          |      |
|         | <b>57.52</b> |   | dB | <b>564.7</b>      | Kbps |



# Example 2 - Link budget

We want to launch a **16U** cubesat around the **Moon**. We will communicate back to Earth using X-band at **8.2GHz** with a patch antenna of **16dB** gain. The ground station antenna has a parabola of **12 meter** of diameter with **3dB** losses. The atmospheric losses are **5dB** and the other losses are negligible.

The data rate at which we would communicate is **5Mbps** with a Bit Error Rate of  **$10^{-6}$  BPSK** modulation. The power of the transmitter is **2W**.

Are we able to have such a communication system?

## Example 2 - Link budget

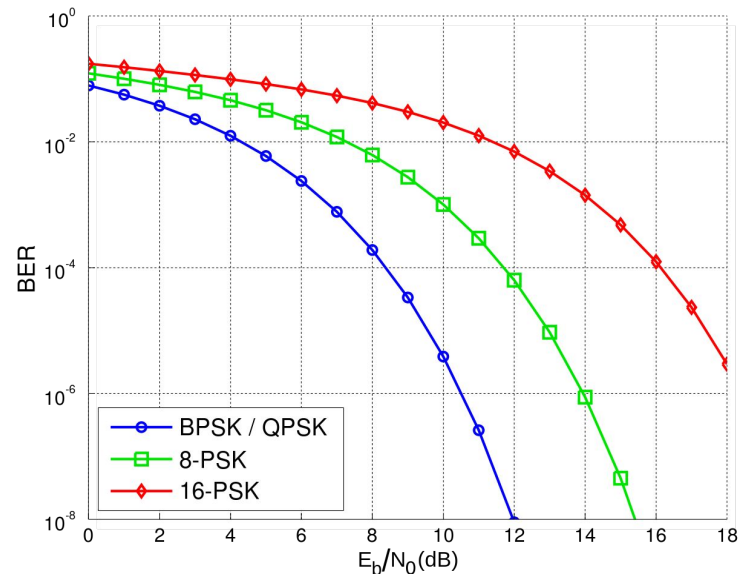
$$G = 20 \log \left( \frac{\pi D f}{c} \right)$$

$$\text{FSL} = 10 \log_{10} \left( \left( \frac{4 \pi d f}{c} \right)^2 \right)$$

$$N [\text{dB}] = 10 \log_{10} \left( \frac{N}{N_{\text{ref}}} \right)$$

$$N = N_{\text{ref}} 10^{\frac{N [\text{dB}]}{10}}$$

$$P_T + G_T - \text{FSL} - L_{\text{ATM}} - L_{\text{tot}} + G_R + 228.6 - E_b N_0 - R > 0$$



# Example 2 - Link Budget

|             |               |   |                 |
|-------------|---------------|---|-----------------|
| Pt          | 3.01          | + | dB              |
| Gt          | 16            | - | dB              |
| FSL         | 222.4         | - | dB              |
| L cable     | 3             | + | dB              |
| Gr          | 64.7          | - | dB <sub>i</sub> |
| T           | 24.8          | + | dB              |
|             | 228.6         | - | dB              |
| EbN0        | 12.0          | - | dB              |
| R           | 67.0          | = | dB              |
| <b>L.B.</b> | <b>-16.86</b> |   | <b>dB</b>       |

&lt; 0 !

# Example 2 - What to do

Link budget is negative so we are not able to communicate at that specific data rate. What can change and how?

1. **Transmitting power ( $P_t$ )**
2. **Spacecraft gain antenna ( $G_t$ )**
3. **Free space losses (FSL)**
4. **Cable losses**
5. **Receiver gain antenna ( $G_r$ )**
6. **Overall system noise**
7. **Data rate**
8. **Bit Error Rate**

# Example 2 - What to do

Link budget is negative so we are not able to communicate at that specific data rate. What can change and how?

1. **Transmitting power ( $P_t$ )**
2. **Spacecraft gain antenna ( $G_t$ )**
3. **Free space losses (FSL)**
4. **Cable losses**
5. **Receiver gain antenna ( $G_r$ )**
6. **Overall system noise**
7. **Data rate (easiest to change, by software)**
8. **Bit Error Rate**

# Example 2 - What to do

Let's put **100Kbps** for the data rate.  
Now the Link Budget (L.B.) is positive,  
hence the communication can happen  
with the quality that we have designed.

|             |             |   |           |
|-------------|-------------|---|-----------|
| Pt          | 3.01        | + | dB        |
| Gt          | 16          | - | dB        |
| FSL         | 222.4       | - | dB        |
| L cable     | 3           | + | dB        |
| Gr          | 64.7        | - | dBi       |
| T           | 24.8        | + | dB        |
|             | 228.6       | - | dB        |
| EbN0        | 12.0        | - | dB        |
| R           | <b>50.0</b> | = | dB        |
| <b>L.B.</b> | <b>0.13</b> |   | <b>dB</b> |

# Example 3 - Long range comms

In the future we want to be able to explore other solar systems in the galaxy. What are the main technological challenges in terms of communicating.

We are able to launch a very small mission to **Proxima Centauri** which is **4.2 light years** away.

Let's make a preliminary design to see if we are able to communicate with the probe from the Earth.

What is the main problem? The main losses that we are going to experience?

# Example 3 - Free space losses

The free space losses will be huge due to the extreme distances.

If we are going to use 8.4GHz as the Voyager 1 spacecraft these are the losses that we are going to have:

4.2 light years =  **$4 \times 10^{13}$  km**

frequency = **8.4 GHz**

$$\text{FSL} = 383 \text{ dB}$$

This is a huge number.

Let's see how much transmitting power we need on the satellite to be able to transmit 10 bit per second to the Earth.



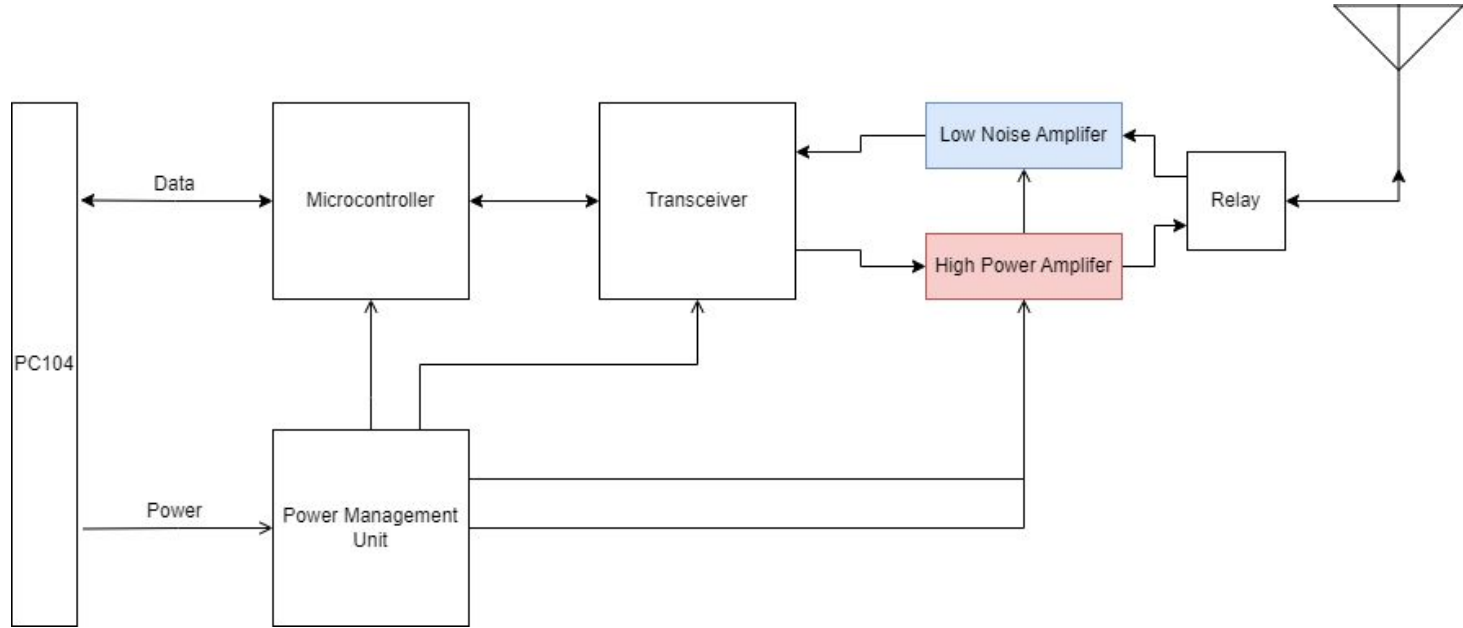
# Example 3 - Power needed

|      |              |   |    |                   |    |
|------|--------------|---|----|-------------------|----|
| FSL  | 383          | + | dB | Free space losses |    |
| EbN0 | 10           | - | dB | EbN0 minimum      |    |
| Gt   | 14           | - | dB | TX Ant. gain      |    |
| Gr   | 62           | + | dB | RX Ant. gain      |    |
| T    | 6            | + | dB | Sys temperature   |    |
| R    | 10           | - | dB | Data rate         |    |
|      | 228.6        | = | dB | Boltzman Const.   |    |
| Pt   | <b>104.4</b> |   | dB | <b>27</b>         | GW |

# Impact on other subsystems

- **Electrical Power System:** the telecommunication system is typically very powerful, especially in the transmitting mode
- **Structural System:** The antennas must operate without any impairment. Could be deployed once in space.
- **Payload System:** The telecommunication system is driven by this system sizing
- **Thermal System:** High temperature can increase the system noise temperature degrading the performance
- **Attitude Determination and Control:** can be used to perform tracking or GNSS

# Transceiver layout example



Questions?